$\boxed{T12}$

$n = 2 \qquad \alpha = 0{,}2$

$H_0: \quad P_1 = \frac{6}{24} \quad P_2 = \frac{6}{24} \quad P_3 = \frac{4}{24} \quad P_4 = \frac{8}{24}$

$H_1: \quad P_1 = \frac{6}{24} \quad P_2 = \frac{6}{24} \quad P_3 = \frac{6}{24} \quad P_4 = \frac{6}{24}$

$H_0$

| P | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/16 | 1/16 | 1/24 | 1/12 |
| 2 | 1/16 | 1/16 | 1/24 | 1/12 |
| 3 | 1/24 | 1/24 | 1/36 | 1/18 |
| 4 | 1/12 | 1/12 | 1/18 | 1/9 |

$H_1$

| P | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/16 | 1/16 | 1/16 | 1/16 |
| 2 | 1/16 | 1/16 | 1/16 | 1/16 |
| 3 | 1/16 | 1/16 | 1/16 | 1/16 |
| 4 | 1/16 | 1/16 | 1/16 | 1/16 |

| $\ell$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | $\boxed{24/16}$ | $12/16$ |
| 2 | 1 | 1 | $\boxed{24/16}$ | $12/16$ |
| 3 | $\boxed{24/16}$ | $\boxed{24/16}$ | $\boxed{36/16}$ | $18/16$ |
| 4 | $12/16$ | $12/16$ | $18/16$ | $9/16$ |

$$P(\ell \geq C \mid H_0) \leq \alpha = 0{,}2$$

$$\frac{1}{36} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} \leq 0{,}2$$

$$\frac{1}{36} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{18} > 0{,}2$$

$$C = \frac{24}{16} = \frac{3}{2}$$

$$G: \ \ell \geq \frac{3}{2}$$

G: выпадение хотя бы одной 3 и не выпадение ни одной 4

$$\alpha_1 = \frac{1}{36} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} + \frac{1}{24} = \frac{1}{36} + \frac{1}{6} = \frac{7}{36}$$

$$W = \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{5}{16}$$

$$\alpha_2 = 1 - W = \frac{11}{16}$$